

Considering the possibility of different effects: Prognostic factors versus relative risk modification/effect modification/interaction/subgroup effect

Users of medical literature often struggle to differentiate between the impact of prognosis (prognostic factors, risk factors, baseline risk) on the magnitude of absolute treatment effects (risk difference) and impact

of subgroup effects (effect modification or subgroup analysis) on the magnitude of risk differences.

Differences in baseline risk—the risk without the intervention of interest— may be associated with patient or disease characteristics (e.g. old versus young). For instance, if the frequency of an adverse outcome is twice as great in the older versus the younger, the absolute effect or risk difference associated with an effective treatment will be twice as great in the older versus the younger.

On the other hand, a subgroup effect occurs when a particular patient or disease characteristic modifies the relative effective of a particular intervention. This means that the relative effect of a therapy differs between one group of patients and another. For instances, the relative risk reduction may be 50% in the elderly and only 25% in younger individuals.

The table below provides an exercise that has helped people distinguish between the impact of differences in baseline risk versus subgroup effect (differences in relative effect).

Table 1. Example of three scenarios presenting hypothetical results for effect modification and prognostic factors

		Outcome in Treatment	Outcome in control (baseline risk)	Relative risk	Absolute effect
Scenario A- most common	Young	10%	20%	$10/20\%=0.5$	$20-10=10\%$
	Old	20%	40%	$20/40\%=0.5$	$40-20=20\%$
Scenario B-unusual	Young	10%	20%	$10/20\%=0.5$	$20-10=10\%$
	Old	40%	40%	$40/40\%=1$	$40-40=0\%$
Scenario C- extremely unusual	Young	10%	20%	$10/20\%=0.5$	$20-10=10\%$
	Old	20%	20%	$20/20\%=1$	$20-20=0\%$

The three scenarios present hypothetical results of three randomized trials (A, B, and C) comparing a treatment group to a control group on a patient-important adverse outcome of interest (e.g. mortality).

Scenario A (different baseline risks, the same relative effects, different absolute effects across subgroups): In scenario A the baseline risk (control event rate) differs between the two groups: 40% in the old, 20% in the young. The relative risk is, however the same in two populations (10%/20% or 0.5 in the young; 20%/40% or 0.5 in the old). The risk difference is 10% in the young (20% - 10%), and 20% in the old (40% - 20%). Thus, this presents an example of constant relative risk, but differences in prognosis influencing the final risk difference.

Scenario B (different baseline risks, different relative effects and absolute effects across subgroups): In scenario B the baseline risk (control event rate) differs between the two groups: 40% in the old, 20% in the young. The relative risk also differs in the two populations (10%/20% or 0.5 in the young; 40%/40% or 1.0 in the old). The risk difference is 10% in the young (20% - 10%), and 0 in the old (40% - 40%). Thus, this presents an example of differences in both relative risk and differences in prognosis influencing the final risk difference.

Scenario C (the same baseline risks, different relative effects, different absolute effects across subgroups): In scenario C the baseline risk (control event rate) is the same, 20%, in the young and old. The relative risk, however, differs (10%/20% or 0.5 in the young; 20%/20% or 1.0 in the old). The risk difference is 10% in the young (20% - 10%), and 0 in the old (20% - 20%). Thus, this presents an example of constant baseline risk, but differences in relative risk influencing the final risk difference.

Scenario A is by far the most common. Prognosis often differs between risk groups such as the young and the old, or the sicker and the less sick. Relative effects are most often similar between these groups with different prognosis.

Scenario B is the next most common, but quite unusual. Groups differ in prognosis but also differ in the magnitude of the treatment effect.

Scenario C in which prognosis is the same in two groups of patients but relative risk differs is extremely unusual.

The following presents an example of Scenario B in which both the baseline risk and the relative risk differ between subgroups. The Study to Prospectively Evaluate Reamed Intramedullary Nails in Patients with Tibial Fractures (SPRINT) trial evaluated the use of alternative regimens for managing tibial fractures.¹ The researchers randomized patients into two groups: one in which the bone marrow was reamed and a large intramedullary nail inserted, and the other receiving unreamed nailing. The authors chose the frequency of reoperation or autodynamization as the primary outcome.

The primary analysis failed to detect a convincing difference between groups, with a relative risk reduction of 10% for reamed versus unreamed nailing (relative risk 0.90; 95% confidence interval 0.71 to 1.15). In a pre-planned subgroup analysis, the researchers hypothesized that reaming would be more beneficial in closed fractures while unreamed nailing would be better for open fractures. For patients with closed tibial fractures, reamed nailing lowered the risk by 33% (relative risk 0.67; 95% confidence interval 0.47 to 0.96), while for patients with open tibial fractures, reamed nailing increased the risk by 27% (relative risk 1.27; 95% confidence interval 0.91 to 1.78) (Table 2).

The statistical interaction test for this subgroup analysis showed a p-value of 0.01. This subgroup analysis was one of five that the researchers investigated. Using the ICEMAN tool,² we can rate the credibility of this subgroup effect as moderate (Table 3).

In regard to baseline risk, patients with open fractures had a greater risk of the primary outcome (29%) than did patients with closed fractures (11%). Therefore, fracture type (open versus closed) serves as both a prognostic factor (open fractures with a worse prognosis) and an effect modifier (reaming better for closed fractures, non-reaming better for open fractures) (Table 2).

Table 2. Example of Scenario B from SPRINT trial

		Outcome in Reamed	Outcome in Unreamed (Baseline risk)	Relative risk	Absolute effect
Scenario B from the SPRINT trial	Closed Fracture	11%	17%	11/17%=0.67	17-11= 6%
	Open Fracture	29%	24%	29/24%=1.27	24-29= -5%

Table 3 Credibility of subgroup analysis for SPRINT trial

CREDIBILITY ASSESSMENT

Essential preliminary considerations to define the possible effect modification of interest

State a single candidate effect modifier (e.g., age or comorbidity):

Was the effect modifier measured before or at randomization? ☒ yes, continue ☐ no, stop here and refer to manual for further instructions

State a single outcome and time-point (e.g., mortality at 1 year follow-up):

State a single effect measure (e.g., relative risk or risk difference):

1: Was the direction of the effect modification correctly hypothesized a priori?

☐ Definitely no	☐ Probably no or unclear	☐ Probably yes	☒ Definitely yes
<i>Clearly post-hoc or results inconsistent with hypothesized direction or biologically very implausible</i>	<i>Vague hypothesis or hypothesized direction unclear</i>	<i>No prior protocol available but unequivocal statement of a priori hypothesis with correct direction of effect modification</i>	<i>Prior protocol available and includes correct specification of direction of effect modification, e.g. based on a biologic rationale</i>
Comment: A priori subgroup hypothesis that was a major focus of the RCT and a factor that was used in stratification of randomization.			

2: Was the effect modification supported by prior evidence?

☐ Inconsistent with prior evidence	☐ Little or no support or unclear	☒ Some support	☐ Strong support
<i>Prior evidence suggested a different direction of effect modification</i>	<i>No prior evidence or consistent with weak or very indirect prior evidence (e.g. animal study at high risk of bias) or unclear</i>	<i>Consistent with more limited or indirect prior evidence (e.g. large observational study, non-significant effect modification in prior RCT, or different population)</i>	<i>Consistent with strong prior evidence directly applicable to the clinical scenario (e.g. significant effect modification in related RCT)</i>
Comment: Some evidence suggests that preserving the endosteal blood supply may be more crucial in open fractures than in closed ones.			

3: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification? (consider irrespective of number of effect modifiers)

☐ Chance a very likely explanation	☐ Chance a likely explanation or unclear	☒ Chance may not explain	☐ Chance an unlikely explanation
<i>Interaction p-value >0.05</i>	<i>Interaction p-value ≤0.05 and >0.01, or no test of interaction reported and not computable</i>	<i>Interaction p-value ≤0.01 and >0.005</i>	<i>Interaction p-value ≤0.005</i>
Comment: The reported p-value for test of interaction is 0.01.			

4: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?

☐ Definitely no	☒ Probably no or unclear	☐ Probably yes	☐ Definitely yes
<i>Explicitly exploratory analysis or large number of effect modifiers tested (e.g. greater than 10) and multiplicity not considered in analysis</i>	<i>No mention of number or 4-10 effect modifiers tested and number not considered in analysis</i>	<i>No protocol available but unequivocal statement of 3 or fewer effect modifiers tested</i>	<i>Protocol available and 3 or fewer effect modifiers tested or number considered in analysis</i>

Comment: The authors tested a total of 6 subgroup hypotheses. All were proposed a priori. The open vs closed fracture subgroup, however, was their primary subgroup of interest.

5: If the effect modifier is a continuous variable, were arbitrary cut points avoided? ☒ not applicable: not continuous

☐ Definitely no	☐ Probably no or unclear	☐ Probably yes	☐ Definitely yes
<i>Analysis based on exploratory cut point (e.g. picking cut point associated with highest interaction p-value)</i>	<i>Analysis based on cut point(s) of unclear origin</i>	<i>Analysis based on pre-specified cut points, e.g. suggested by prior RCT</i>	<i>Analysis based on the full continuum, e.g. assuming a linear or logarithmic relationship</i>

Comment: The subgroup hypothesis is binary in nature. Thus, this domain is not applicable.

6 Optional: Are there any additional considerations that may increase or decrease credibility? (manual section 2.6)

☐ yes, probably decrease ☒ yes, probably increase

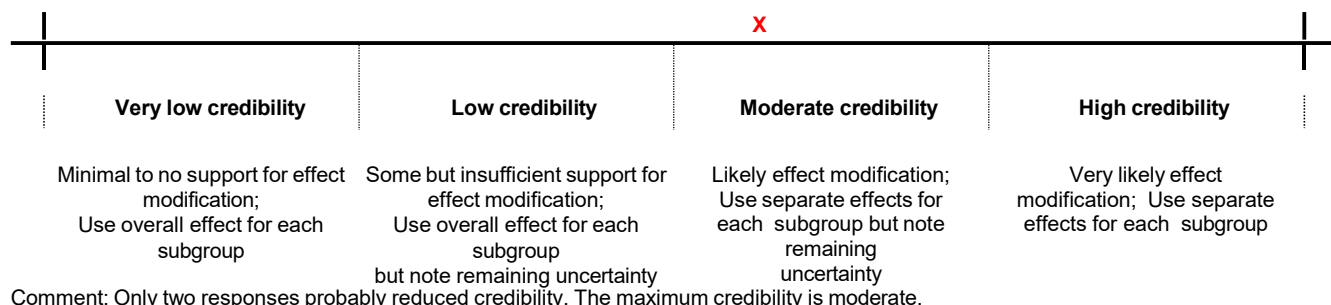
Comment: The investigators stratified their randomization based on this factor. This ensured adequate representation of open and closed fraction in both the intervention and control arm.

7: How would you rate the overall credibility of the proposed effect modification?

The overall rating should be driven by the items that decrease credibility. The following provides a sensible strategy:

- All responses definitely or probably reduced credibility or unclear very low
- Two or more responses definitely reduced credibility maximum usually low even if all other responses satisfy credibility criteria
- One response definitely reduced credibility maximum usually moderate even if all other responses satisfy credibility criteria
- Two responses probably reduced credibility maximum usually moderate even if all other responses satisfy credibility criteria
- No response options definitely or probably reduced credibility high very

likely Place a mark on the continuous line (e.g. hit "x" in electronic version)



References

1. Bhandari M, Guyatt G, Tornetta P, 3rd, Schemitsch EH, Swiontkowski M, Sanders D, Walter SD. Randomized trial of reamed and unreamed intramedullary nailing of tibial shaft fractures. *J Bone Joint Surg Am* 2008;90(12):2567-78. doi: 10.2106/jbjs.G.01694
2. Schandelmaier S, Briel M, Varadhan R, Schmid CH, Devasenapathy N, Hayward RA, Gagnier J, Borenstein M, van der Heijden G, Dahabreh IJ, Sun X, Sauerbrei W, Walsh M, Ioannidis JPA, Thabane L, Guyatt GH. Development of the Instrument to assess the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials and meta-analyses. *Cmaj* 2020;192(32):E901-e06. doi: 10.1503/cmaj.200077